

ARAVINDSELVAN C

RF & ANTENNA ENGINEERING | ECE UNDERGRADUATE, CLASS OF 2027

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ECE undergraduate and licensed HAM radio operator (VU37AE), with internship experience in SDR signal reception and a team dipole-antenna design and testing project. A childhood interest in space has grown through ISRO learning activities and practical RF work. Seeking early-career opportunities in RF and antenna engineering.

EDUCATION

B.E. Electronics and Communication Engineering

Sri Sairam Engineering College, Chennai | Sep 2023 - May 2027 (expected)

CGPA: 6.95/10

RELEVANT INTERNSHIP

Advanced SDR for LEO Satellite Signal Acquisition and Ham Radio Signals

Centre of Excellence - Space Technology, Sri Sairam Engineering College

15-day internship | 16 - 30 June 2026

- Participated in practical SDR sessions using RTL-SDR, SDR# and SDR++ for FM reception and RF spectrum observation; used GPredict for LEO satellite tracking and pass prediction.
- Undertook ADS-B reception practicals using RTL1090/Dump1090 and NOAA signal-reception exercises; studied satellite communication and amateur-radio operating practices.
- Gained laboratory exposure to Keysight FieldFox RF measurements, including VSWR and return loss, through a team antenna project.

RF PROJECT

435 MHz Half-Wave Dipole Antenna | Three-member internship team

- Designed a half-wave dipole as part of a three-member team project involving fabrication and RF testing; the documented design used 16.4 cm copper arms and an approximately 0.7 cm feed gap.
- The team investigated mounting and alignment effects, replaced the initial PVC support with a wooden support, and repeated FieldFox measurements after checking the feed connection.
- Documented antenna construction and RF test results in the joint internship report; the included analyzer trace shows 21.9 dB return loss at approximately 426.79 MHz, near the 435 MHz design target.

TECHNICAL FOUNDATIONS & TOOLS

RF / communications: Dipole antennas, wavelength-based sizing, impedance-matching concepts, VSWR, return loss, SDR reception, LEO satellite tracking.

Internship tools: RTL-SDR, SDR#, SDR++, GPredict, RTL1090/Dump1090; Keysight FieldFox (laboratory exposure).

Programming: Python, C, C++ | **Supporting tools:** Git, GitHub, Linux, Jupyter Notebook.

SPACE LEARNING & AMATEUR RADIO

- **Licensed amateur (HAM) radio operator:** VU37AE; ongoing interest in radio communication and space technology.
- **IIRS, ISRO:** Participated in the Online Summer School on Usefulness of Remote Sensing & GIS for Environmental Studies, 26 - 30 July 2021 (10 hours).
- **ISRO Cyberspace Competitions 2020:** Participated in the English essay competition, Class 9 & 10 category; certificate dated 25 September 2020.

ADDITIONAL PROJECT LEADERSHIP

AirTon | Technical Lead: Student-team ocular-screening research prototype. Worked on the machine-learning component using AI tools.